# Kush Mehta

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## Education

Graduate Engineering Data Science student with hands-on experience across data engineering, analytics, and applied machine learning in academic and industry environments. Proven ability to build end-to-end data pipelines, transform large datasets, and deliver insights through scalable workflows, dashboards, and production-grade analytics systems.

## Professional Experience

**University of Houston** — **Jan 2025 – Present**
*Research Assistant – Data Engineering*
- Designed and built large-scale ETL pipelines using **Apache Airflow** (DAGs, task dependencies, scheduling) to automate ingestion and transformation of 10M+ structured records in Snowflake.
- Optimized **complex SQL stored procedures, CTEs, and window functions** to support large-scale analytical queries, reducing **end-to-end ETL runtime by 40%** across **multi-terabyte Snowflake datasets**.
- Implemented **Power BI** dashboards with DAX-based measures and row-level aggregations to track pipeline health, volumes, and downstream analytics consumption, improving visibility into system performance.

**Sahu Technologies** — **Aug 2023 – May 2024**
*Data Engineer Intern*
- Constructed **AWS Glue**–based ingestion pipelines to process **3M+ high-volume event** and customer interaction records, enabling analytics on energy service usage, demand patterns, and operational behavior.
- Modeled time-partitioned analytical datasets in **Databricks using Delta Lake and Delta Live Tables**, applying dimensional schemas to reduce aggregation query latency by 45% across time-series–driven performance.
- Developed Alteryx BI workflows and dashboards to track service KPIs, usage trends, and operational reliability.

**Pheme Software Pvt. Ltd. – IBM Partner** — **Dec 2022 – Mar 2023**
*Data Analyst Intern*
- Evaluated machine learning models in Python, including Random Forest classifiers for issue severity, and Isolation Forest models for anomaly detection, reducing operational triage time by 30% across client projects.
- Integrated **Apache Kafka**–based streaming workflows to ingest and process high-volume log and event data in near real time, enabling scalable feature generation for downstream analytics and ML pipelines.
- Containerized ML services with **Docker** and deployed **CI/CD pipelines on AWS Lambda**, reducing deployment cycles.

## Technical Skills

- **Programming Languages & Tools** - Python, Java, Flask, Streamlit, XML, R, SQL, SAS, MATLAB, Git, MS Excel
- **Frameworks & Visualization** - Numpy, Pandas, Matplotlib, NLTK, PyTorch, TensorFlow, Power BI, Tableau
- **Data Engineering & Deployment** - Airflow, DBT, PySpark, Docker, Kubernetes, GitHub Actions, Jenkins
- **Databases & Visualization** – PostgreSQL, AWS RDS, MongoDB Atlas, Azure Cosmos DB, MySQL, Power BI, Tableau
- **Certifications** - IBM Advanced Certificate in AI & ML, AWS NLP for ML, AWS Cloud Foundations
- **Relevant Skills** - Analytical Thinking, Communication, Team Collaboration, Project Management, Continuous Learning

## Education

**University of Houston** — **Aug 2024 – May 2026**
*Master of Science in Engineering Data Science* — *GPA: 3.83/4.0*

**University of Mumbai** — **Aug 2020 – May 2024**
*Bachelor of Engineering in Computer Engineering & Honors in AIML* — *GPA: 3.87/4.0*

## Academic Projects

**BroadVail RevPAR Intelligence Platform – Broadvail Capital** | *Python, Streamlit, XGBoost, LightGBM, SHAP*
- Engineered a scalable revenue forecasting pipeline over 12,800+ multifamily property records using XGBoost, LightGBM, achieving 70.6% $R^2$ and reducing RMSE by 18.3% compared to baseline models.
- Automated reporting with LLM summary and SHAP-based explainability, delivering investment insights via dashboards.

**Predictive Maintenance System for Aircraft Engines** | *Python, TensorFlow, Scikit-learn, NumPy, Pandas*
- Applied predictive maintenance models in Python (Random Forest, XGBoost) on 2M+ IoT sensor records in AWS S3.
- Queried data with SQL on Redshift, applying time-series methods to identify failure patterns, surge precision by 18%.

**Real-Time Construction Equipment Utilization Dashboard** | *Kafka, Spark, MongoDB, BigQuery, Tableau*
- Built a real-time streaming pipeline using Spark to ingest high-frequency construction equipment telemetry and GPS events, storing analytical data in BigQuery to enable near-real-time utilization tracking across sites.
- Produced dashboards using Tableau to visualize KPIs and idle-time patterns, improving equipment utilization by 25%.

**EEG-Based Motor Movement Classification – Neurotech** | *Python, TensorFlow, PyTorch, NLTK, CUDA*
- Fabricated an EEG signal processing pipeline on a 64-channel dataset (86 participants) for feature extraction.
- Trained Random Forest, SVM, and CNN models to classify hand vs foot movements, achieving ~78% accuracy.